

Trigonometric functions, their properties and graphs

The functions that repeat — and the periodic processes they describe.

LESSON 71–72

2 × 40 MIN

ALGEBRA 10, §4.1

P1 · 5.2

LESSON CLOCK

14'

24'

22'

20'

16'

8'

The unit-circle definition	14 min
The three graphs	24 min
Properties	22 min
Periodic processes	20 min
Practice	16 min
Homework	8 min

LEARNING OBJECTIVES

- Define sine, cosine and tangent on the unit circle for any angle.
- Sketch $y = \sin x$, $y = \cos x$ and $y = \tan x$.
- State the period, range, zeros and symmetry of each.
- Model a periodic process with a trigonometric function.

TEXTBOOK

National	pp. 301–312
Cambridge	Pure Mathematics 1, pp. 96–104
Workbook	P1 Exercise 5B

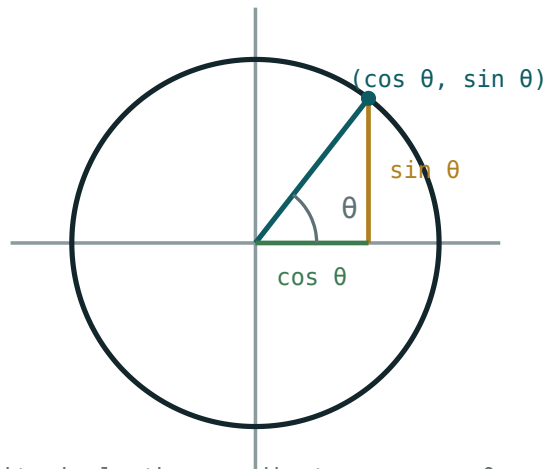
01

Explanation

The unit-circle definition

Take the point P on the unit circle at angle θ from the positive x -axis, measured anticlockwise. Then by definition

$$\cos \theta = x\text{-coordinate of } P \quad \sin \theta = y\text{-coordinate of } P$$



on the unit circle the coordinates are $\cos \theta$ and $\sin \theta$

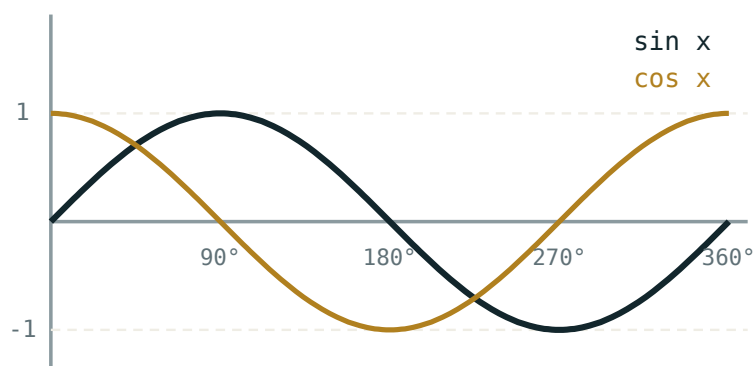
The coordinates of the point are the cosine and the sine. Nothing else is needed.

WHY THIS REPLACES THE RIGHT TRIANGLE

The triangle definition works only for $0^\circ < \theta < 90^\circ$. The circle definition works for every angle — negative, obtuse, reflex, or beyond a full turn — and it makes the signs in each quadrant obvious from the coordinates.

Because P returns to the same place after a full turn, both functions repeat with period 360° . The tangent is $\tan \theta = \frac{\sin \theta}{\cos \theta}$, undefined where $\cos \theta = 0$.

The three graphs



period 360° , range -1 to 1

One period of each. The cosine is the sine shifted left by 90° .

	<i>SIN X</i>	<i>COS X</i>	<i>TAN X</i>
period	360°	360°	180°
domain	$\mathbb{R}$	$\mathbb{R}$	$x \neq 90^\circ + 180^\circ k$
range	$[-1, 1]$	$[-1, 1]$	$\mathbb{R}$
zeros	$180^\circ k$	$90^\circ + 180^\circ k$	$180^\circ k$
parity	odd	even	odd
maximum	1 at 90°	1 at 0°	none

THE TANGENT HAS ASYMPTOTES

At $90^\circ, 270^\circ, \dots$ the cosine is zero and the tangent is undefined. Its graph is a chain of identical branches, each rising from $-\infty$ to $+\infty$ over an interval of 180° .

Properties, and what they are for

$$\sin(-x) = -\sin x \cdot \cos(-x) = \cos x \cdot \tan(-x) = -\tan x$$

$$\sin^2x + \cos^2x = 1$$

The last is Pythagoras on the unit circle: the point $(\cos \theta, \sin \theta)$ is at distance 1 from the origin. Every trigonometric identity of Quarter IV is built from it.

QUADRANT	<i>SIN</i>	<i>COS</i>	<i>TAN</i>
I ($0\text{--}90^\circ$)	+	+	+
II ($90\text{--}180^\circ$)	+	−	−
III ($180\text{--}270^\circ$)	−	−	+
IV ($270\text{--}360^\circ$)	−	+	−

Periodic processes

$$y = A \sin(Bx) + C$$

A is the **amplitude** — half the distance from lowest to highest. C is the middle value.

The period is $\frac{360^\circ}{B}$.

PROCESS	PERIOD	MODEL
daylight hours in Tashkent	365 days	$12 + 3 \sin(\dots)$
tides	≈ 12.4 hours	$mean + amplitude \cdot \sin(\dots)$
mains alternating current	0.02 s	$220\sqrt{2} \sin(100\pi t)$
a Ferris wheel	one revolution	$centre\ height + radius \cdot \sin(\dots)$

02

Worked examples

EXAMPLE 1 State the period, amplitude and range of $y = 3 \sin(2x) + 1$.

1

Amplitude $|A| = 3$.

2

Period $\frac{360^\circ}{2} = 180^\circ$.

3

Middle value 1.

4

Range 1 ± 3 .

ANSWER

Period 180° , amplitude 3, range $[-2, 4]$

EXAMPLE 2 Without a calculator, give the sign of $\sin 200^\circ$, $\cos 200^\circ$ and $\tan 200^\circ$.

- ① 200° is in quadrant III.
- ② There sine and cosine are both negative.
- ③ Their quotient is positive.

ANSWER

$-$, $-$, $+$

EXAMPLE 3 Daylight in Tashkent varies from 9.3 h to 15.1 h. Find A and C.

- ① $C = \frac{15.1 + 9.3}{2} = 12.2$

The middle.

- ② $A = \frac{15.1 - 9.3}{2} = 2.9$

Half the swing.

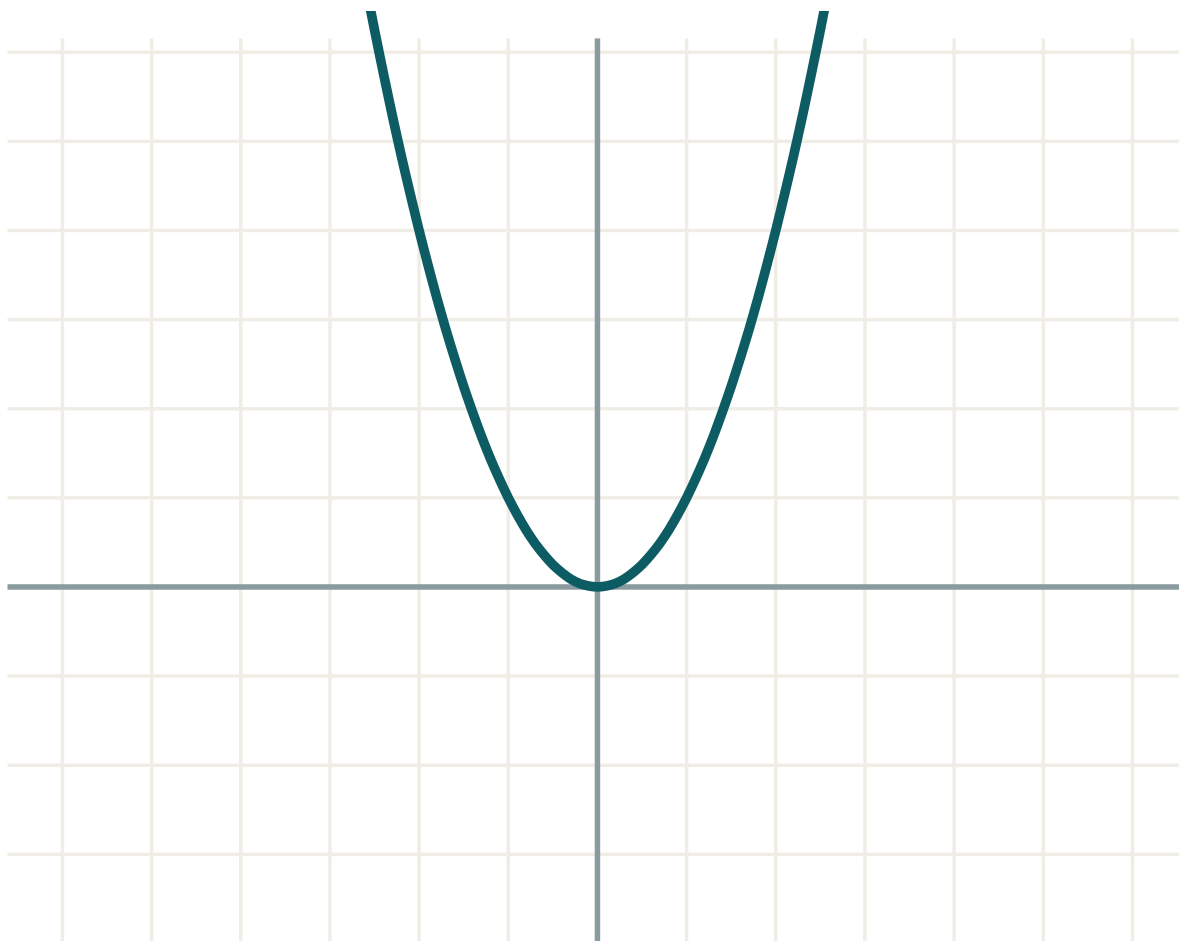
- ③ Period 365 days.

ANSWER

$A = 2.9$, $C = 12.2$ hours

03 Interactive model

Trace a point round the unit circle and plot its height against the angle.

A sine wave**a** 1**b** 0**c** 0equation **y = f(x)****04 Terminology**

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Trigonometric function	Trigonometrik funksiya	Тригонометрическая функция
Unit circle	Birlik aylana	Единичная окружность
Period	Davr	Период
Amplitude	Amplituda	Амплитуда
Zero of a function	Funksiya nuli	Нуль функции
Even function	Juft funksiya	Чётная функция
Odd function	Toq funksiya	Нечётная функция
Asymptote	Asimptota	Асимптота
Periodic process	Davriy jarayon	Периодический процесс

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

The period of $\sin x$ is:

The period of $\tan x$ is:

$\cos(-x)$ equals:

The range of $\cos x$ is:

$\mathbb{R}$

$[-1, 1]$

$[0, 1]$

$x \neq 0$

The amplitude of $y = 3 \sin 2x$ is:

2

3

6

1.5

The period of $y = 3 \sin 2x$ is:

180°

360°

720°

120°

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. Period of $\sin x$

ANSWER 360°

2. Period of $\tan x$

ANSWER 180°

3. Range of $\sin x$

ANSWER $[-1, 1]$

4. Zeros of $\sin x$

ANSWER $180^\circ k$

5. Is $\cos x$ even or odd?

ANSWER even

6. Is $\sin x$ even or odd?

ANSWER odd

7. $\sin 0^\circ$

ANSWER 0

Medium · the standard question

7 PROBLEMS

1. Period and amplitude of $y = 3 \sin 2x$

ANSWER $180^\circ, 3$

2. Range of $y = 3 \sin 2x + 1$

ANSWER $[-2, 4]$

3. Signs of $\sin$, $\cos$, $\tan$ at 200°

ANSWER $-, -, +$

4. Signs at 130°

ANSWER $+, -, -$

5. Period of $y = \cos(3x)$

ANSWER 120°

6. Amplitude of $y = -5 \cos x$

ANSWER 5

7. Where is $\tan x$ undefined?

ANSWER $x = 90^\circ + 180^\circ k$

Hard · stretch and reason

7 PROBLEMS

1. Daylight 9.3 to 15.1 h: find A, C

ANSWER 2.9, 12.2

2. A Ferris wheel of radius 20 m, centre 22 m up, one turn in 4 min: model the height

ANSWER $22 + 20 \sin(90t^\circ)$

3. Range of $y = 2 - 4 \cos x$

ANSWER $[-2, 6]$

4. Period of $y = \sin\left(\frac{x}{2}\right)$

ANSWER 720°

5. Solve $\sin x = 0$ on $[0^\circ, 720^\circ]$

ANSWER 0, 180, 360, 540, 720

6. Show $\sin^2 x + \cos^2 x = 1$ from the unit circle

ANSWER It is Pythagoras on the radius

7. Sketch $y = \tan x$ on $[-180^\circ, 180^\circ]$ with its asymptotes

ANSWER Asymptotes at $\pm 90^\circ$

07 Homework

Homework — 5 tasks

Every sketch must be labelled with the period and, for the tangent, the asymptotes.

- Sketch $y = \sin x$ and $y = \cos x$ on $[0^\circ, 720^\circ]$ on one set of axes.
- State the period, amplitude and range of $y = 4 \sin(3x) - 2$.
- Give the signs of the three functions at 110° , 250° and 320° .
- A tide varies from 1.2 m to 4.8 m with period 12.4 hours. Find the amplitude and the mean level.

5. Sketch $y = \tan x$ on $[-180^\circ, 180^\circ]$, marking the asymptotes and the zeros.